

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**End Semester Examination – Winter 2018****Course: B. Tech in Civil Engineering****Sem: III****Subject Name: Surveying-I****Subject Code: BTCVC304****Max Marks: 60****Date: 07/12/2018****Duration: 3 Hr.****Instructions to the Students:**

1. Solve **ANY FIVE** questions out of the following.
2. The level question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks	
Q.1 Solve Any Two of the following.		12	
A) Give the conventional symbols for the following:	CO2	06	
1) North line , Chain line			
2) Main station, Traverse station			
3) River, Canal			
4) Road Bridge, Boundary line			
5) Jungle, Wire fencing			
6) Railway line (double), Open well			
B) What is Well-Conditioned triangle? Explain clearly why it is preferred instead of an Ill-conditioned triangle.	CO1	06	
C) What is Cross Staff? List down the types of Cross staff. Describe the construction and use of an Open Cross staff with a sketch.	CO1	06	
Q.2 Solve Any Two of the following.		12	
A) i) The fore bearings of the lines AB, BC, CD, DE are $45^{\circ}30'$, $120^{\circ}15'$, $200^{\circ}30'$, $280^{\circ}45'$ respectively. Find angles $\angle B$, $\angle C$, $\angle D$.	CO1	06	
ii) A traverse is done by three stations A, B and C in clockwise order in the form of equilateral triangle. If the bearing of AB is $80^{\circ}30'$, find the bearings of the other sides.			
B) The following are the observed bearings of the lines of a traverse ABCDEA with a compass in a place where local attraction was suspected.	CO1	06	
Line	FB	BB	Find correct bearings of the lines by calculating interior angles.
AB	$191^{\circ}00'$	$13^{\circ}00'$	
BC	$39^{\circ}30'$	$222^{\circ}30'$	
CD	$22^{\circ}15'$	$200^{\circ}30'$	
DE	$242^{\circ}45'$	$62^{\circ}45'$	
EA	$330^{\circ}15'$	$147^{\circ}00'$	

- C) Followings are the bearings observed while traversing with a compass, an area where local attraction was suspected. Find the correct bearings of the lines and also the true bearings, if the magnetic declination is 09° W. **CO1 06**

Line	FB	BB
AB	$60^{\circ}00'$	$240^{\circ}00'$
BC	$139^{\circ}30'$	$317^{\circ}00'$
CD	$215^{\circ}15'$	$36^{\circ}30'$
DE	$208^{\circ}00'$	$29^{\circ}00'$
EA	$318^{\circ}30'$	$138^{\circ}45'$

Q. 3 Solve Any One of the following.

- A) Write short note on: **CO1 12**
- | | |
|--------------------------|--|
| i) Reconnaissance Survey | iv) Layout |
| ii) Preliminary Survey | v) Difference between Summit curve & Valley curve |
| iii) Location Survey | vi) Factors responsible for selection of good alignment for a road |
- B) Explain: **CO1 12**
- | | |
|-------------------------|--|
| i) Curve ranging | iii) Objectives behind provision of curves |
| ii) Definition of curve | iv) Construction survey for Waterways |

Q.4 Solve Any Two of the following.

- A) What is the Principle of Plane Tabling? Explain all accessories of Plane table with neat figures. **CO2 06**
- B) List down the methods of plane tabling. Explain procedure of any two methods with neat figures. **CO2 06**
- C) Write down the advantages and disadvantages of plane tabling. **CO2 06**

Q. 5 Solve Any Two of the following.

- A) i) Draw neat labeled diagram of Dumpy level. **CO3 06**
 ii) Write down the object and uses of leveling.
 iii) Write down definitions of : 1) Line of Collimation 2) Change Point
- B) i) An observer standing on the deck of a ship sees a light house. The top of light house is 35m above the sea level and the height of the observer is 6m above sea level. Find the distance between observer and light house. **CO1, CO3 06**
 ii) In leveling between two points A and B on the opposite sides of a river, the level was set up near A and the staff readings on A and B were 2.645m and 3.230m respectively. The level was then moved and set up near B, the respective readings on A and B was 1.085m and 1.665m. Find the true difference of level between A and B

iii) A bench mark with reduced level 155.305m has been established at the floor of a room. It is required to find out the RL of the underside of the roof(R) of the room using spirit leveling. The back sight to the bench mark has been observed as 0.575m (staff held inverted). Calculate the RL(m) of R.

- C) Define the term contour line. Explain the characteristics of contour lines with neat figures.

CO3 06

Q.6 Solve Any One of the following.

- A) i) Draw neat labeled diagram of Transit Theodolite.
 ii) A flag post of height 2m was erected on top of a building. Find the RL of the top of the flag post, if the vertical angles to the bottom and top of it were measured using Theodolite as 7° and 10° respectively. A staff reading of 1.245m was taken over a bench mark of +150.000m with vertical angle of $0^\circ 0' 0''$.
 iii) Explain Balancing of traverse and rules of distributing errors.

- B) The length and bearing of lines of a closed traverse ABCDA are as under.

Line	Length	WCB	Calculate latitude and departure. Apply correction to
AB	250	130°	latitude and departure. Find out closing error. Use
BC	600	42°	Bowditch rule.
CD	100	317°	
DA	635.46	$235^\circ 40'$	

*** End ***

12

12

12

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**End Semester Examination – May 2019****Second Year (III Semester), B. Tech. in Civil Engineering****Time: 03 Hours****Sub: Surveying - I (BTCVC304)****Max Marks: 60****INSTRUCTIONS TO THE STUDENTS**

1. Attempt any five questions from the given six questions.
2. Illustrate your answers with neat sketches, diagrams etc. where ever necessary.
3. Necessary data is given in the respective questions. If such data is not given, it means that the knowledge of that data is a part of the examination.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly.

(Marks)**Q.1. Solve any three from the following:**

- (a) What are the different obstacles in chaining operations? Explain any one method of overcoming the obstacle? (4)
- (b) Explain: Construction and use of 20 m chain (with sketch). (4)
- (c) Write a short note on mistakes in chaining. (4)
- (d) A 30 m chain was tested before the commencement of the day's work and was found to be correct. After chaining a distance of 100 chains it was found to be 5 cm too short. At the end of day's work after chaining a total distance of 180 chains, the chain was found to be 10 cm too long. What was the true distance chained? (4)

- Q.2. (a) Define the following terms used in compass surveying: Dip of needle, Local attraction, and True meridian. (6)
- (b) Compute the interior angles of the traverse PQRST if the bearings of the side are: (6)

Side	Fore Bearing	Back Bearing
PQ	107° 15' 00"	287° 15' 00"
QR	22° 00' 00"	202° 00' 00"
RS	231° 30' 00"	101° 30' 00"
ST	189° 15' 00"	9° 15' 00"
TP	124° 45' 00"	304° 45' 00"

- Q.3. (a) Explain in brief the intersection method of plane tabling. (6)
- (b) List the minor instruments used in plane table surveying. (6)

- Q.4. (a) A level is set up at station P. The reading on the staff when held at point A (6)
360 m away from O, is 2.150 and is 3.895 when held at other point B,
550 m away. Find the true difference of level between A and B.
- (b) Write the fundamental characteristics of contour lines? What are the uses (6)
of contour maps?

- Q.5. (a) List the parts of a Theodolite. (12)

Following deflection angles were measured in running a traverse from A to G.

Station	Deflection angle	Station	Deflection angle
B	23° 47' R	E	15° 38' R
C	18° 19' L	F	10° 12' L
D	37° 20' R		

If the true bearing of AB is N 63° 18' E, find the true bearings of the other sides.

or

The following is the closed theodolite traverse. Balance the traverse and (12)
calculate the independent coordinates in the form of Gale's Table.

Line	Length (m)	Angle	Remarks
AB	1618	A 55° 20'	Bearing of line
BC	1050	B 104° 47'	AB = 37° 18'
CD	1725	C 54° 17'	
DA	485	D 145° 23'	

- Q.6. (a) Write a short note on route surveying for highways. (6)
- (b) Write a short note on mine surveying. (6)

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DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE – RAIGAD -402 103
Winter End Semester Examination – December 2019

Branch: Civil Engineering

Sem.:- III

Subject- Surveying-I

Subject Code (BTCVC304)

Marks: 60

Date:- 17/12/2019

Time:- 3 Hr.

Instructions to the Students:

1. Each question carries 12marks.
2. Solve **Any Five** questions of the following.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly.

Q.1 Solve Any Two of the following.

12

- A) Two stations A and B are not intervisible due to rising ground between them. Explain with a neat sketch how the line AB can be ranged?
- B) What is mean by offset? List the instruments used for setting the offsets. Describe the construction of open cross staff with neat sketch.
- C) An old map was plotted to a scale of 40m to 1cm. Over the years, this map has been shrinking, and a line originally 200mm long is only 195mm long at present. Again the 20m chain was 5cm too long if the present area of the map measured by the planimeter is 125.50cm², find the true area of the land surveyed.

Q.2 Solve Any Two of the following.

12

- A) The following bearings were observed where local attraction was suspected. Calculate the actual bearings.

Line	FB	BB
AB	40°30'SW	41°15'NE
BC	80°45'SW	79°30'NE
CD	19°30'NE	20°00'SW
DA	80°00'SE	80°00'NW

- B) Define the following terms:

- a) Whole circle bearing
- b) Local attraction
- c) Closing error

- C) In an anticlockwise traverse ABCA all sides were equal. Magnetic fore bearing of BC was obtained to be $15^{\circ}30'$. The bearing of sun was observed to be $184^{\circ}30'$ at local noon with a prismatic compass. Calculate the magnetic bearing and true bearing of all the sides of a traverse.

Q.3 Solve Any Two of the following.

12

- A) What is the principle of plane table surveying? List all accessories of Plane Table with a neat sketch and state use of each.
- B) Explain Methods of plan table
- i) Intersection Method
 - ii) Radiation Method
- C) How will you set plane table? Explain the procedure.

Q.4 Solve Any Two of the following.

12

- A) Define contour. Explain the characteristics of contour with neat sketch.
- B) During a fly levelling operation the following observations were made
- BS : 0.650, 2.155, 1.405, 2.655, 2.435m
- FS : 2.455, 1.305, 0.555, 2.405m
- The first back sight was taken on a BM of RL 90.500m. From the last back sight, it is required to set four pegs each at a distance of 30m on a falling gradient of 1 in 100. Calculate the RLs of these four pegs.
- C) A levelling instrument was set up exactly mid-way between two pegs A and B, 100m apart. The staff reading on A and B were 1.875 and 1.790 respectively. The instrument was then set up at a distance of 10m from A on the line AB. The respective staff readings were 1.630 and 1.560.
- Calculate the correct staff reading on A and B when the line of collimation was exactly horizontal.

Q.5 Solve Any Two of the following.

12

- A) i) What are the different checks in closed traverse in theodolite traversing.
- ii) How will you measure the horizontal angle by repetition method? Explain with an example.

B) The following observations were taken from stations P and Q. Calculate the length and bearing of AB, and also the angles PAB and QBA.

Line	Length(m)	Bearings
PA	125.0	60°30'SW
PQ	200.0	30°30'NE
QB	150.5	50°15'NW

C) The following records are obtained in a traverse survey, where the length and bearing of the last line were not recorded. Compute the length and bearing of line DA.

Line	Length (m)	Bearing
AB	75.50	30°24'
BC	180.50	110°36'
CD	60.25	210°30'
DA	?	?

Q.6 Solve the following.

12

A) Write short note on:

- Factors necessary for the selection of good alignment for a road.
- Setting out a building with an example.

B) Define curve, how it is designated? What are the different classification of a curve explain by sketch.

=====Paper End=====

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Winter Examination – 2022

Course: B. Tech.

Branch :CIVIL

Semester :III

Subject Code & Name:BTCVC305 SURVEYING

Max Marks: 60

Date:

Duration: 3 Hr.

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

(Level/CO) Marks

- Q. 1 Solve Any Two of the following. 12**
- A) Write short note on Metric Chain. 6
 - B) Write short note on: types of cross staff 6
 - C) Write short note on: Advantages & disadvantages of Chain. 6
- Q.2 Solve Any Two of the following. 12**
- A) Explain Local attraction? How it is detected? 6
 - B) Write instruments used in Plane table Surveying. 6
 - C) Convert following reduce bearings to Whole circle bearing. 6
- 1) $S70^{\circ} 40' W$ 2) $N 55^{\circ} W$ 3) $S 24^{\circ} 15' E$
- Q. 3 Solve Any Two of the following. 12**
- A) What are the instruments used in levelling. 6
 - B) Define bench mark. Explain types of bench mark in brief. 6
 - C) Write note on Characteristics and uses of contours. 6
- Q.4 Solve Any Two of the following. 12**
- A) Enlist various fundamental lines of a transit (20") theodolite & explain the relation between them. 6
 - B) Explain measurement of horizontal angle by repetition method. 6
 - C) Write short note on: Gales traverse table 6
- Q. 5 Solve Any Two of the following. 12**
- A) Explain Preliminary Survey. 6
 - B) Write short note on Mine Surveying. 6
 - C) Write short note on Instruments used during tunnel Surveying. 6

***** End *****

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: B. Tech. Branch : Civil Engg Semester :III Subject Code & Name: Surveying –I (BTCVC304) Max Marks: 60 Date: Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following. (This is just a sample instruction)		12
A)	Explain chaining ,ranging ,offsetting in details.	CO1	6
B)	What is meant by well conditioned and ill conditioned triangles?	CO1	6
C)	Explain principle of chain surveying with classification of chaining.	CO1	6
Q.2	Solve Any Two of the following. (This is just a sample instruction)		12
A)	The Bearing of line AB is $153^{\circ}30'$ and angle ABC is $135^{\circ}40'$.What is bearing of BC.	CO1	6
B)	Explain with diagram prismatic compass and surveyor compass in detail.	CO1	6
C)	Define- 1. Whole circle bearing 2. Reduced bearing 3. Fore bearing 4. Back bearing	CO1	6
Q. 3	Solve Any Two of the following. (This is just a sample instruction)		12
A)	Explain the intersection method & resection method of plane table survey.	CO2	6
B)	State any six advantages and disadvantages of plane table survey.	CO2	6
C)	Explain principle of plane table surveying and write down instrument used in this survey.	CO2	6
Q.4	Solve Any Two of the following. (This is just a sample instruction)		12
A)	Explain types of bench marks.	CO2	6
B)	Write characteristics and uses of contours .	CO2	6
C)	Explain in detail types of planimeter.	CO2	6
Q. 5	Solve Any Two of the following. (This is just a sample instruction)		12
A)	Explain types of engineering survey in detail.	CO3	6
B)	Explain Mine surveying and Route surveying in detail.	CO3	6

C)	Explain temporary and permanent adjustment of theodolite.	CO3	6
	*** End ***		

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Course: B. Tech.

Branch :Civil

Semester :III

Subject Code & Name: BTCVC 305 Surveying

Max Marks: 60

Date:

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

- A) What is meant by combined corrections in leveling? Derive its equation. CO1 6
- B) Differentiate between radiation & intersection. CO2 6
- C) A dumpy level was setup & leveled with its eyepiece over a peg C. The height from top of the center of eyepiece was measured & found to be 1.246 m. The staff reading on peg D was 0.892m. The instrument was then moved & set up at D. The height of eyepiece above D was 1.362m. The reading on staff held on peg C was 1.720 m. Determine the true Reduce level of the peg D if that of peg C was 150m. CO3 6

Q.2 Solve Any Two of the following.

- A) A contour plan is to be prepared for building site. State the factors you will consider for the choice of contour interval CO1 6
- B) What are various methods of contouring? Explain any one with neat sketch. CO2 6
- C) The following observations were made during testing of dumpy level. CO3 6

Instrument at	Staff reading at	
	A	B
A	1.725	2.245
B	2.145	3.045

What should be R.L. of B if R.L. of A is 450.00m?

Q. 3 Solve Any One of the following.

- A) What is zero circle? What is need of finding the area of zero circle? CO1 6
- B) Write note on Sensitivity of bubble tube. CO2 6
- C) What is orientation of plane table? Explain any one method of it. CO3 6

Q.4 Solve Any Two of the following.

- | | | |
|--|------------|----------|
| A) What do you understand by the term Balancing the traverse? State various rules available for this process. | CO1 | 6 |
| B) Explain the procedure for carrying out preliminary survey for proposed irrigation canal. | CO2 | 6 |
| C) Explain spire test for adjusting the Transit theodolite. | CO3 | 6 |

Q. 5 Solve Any One of the following.

- | | | |
|---|------------|----------|
| A) Explain with neat sketch the method of transferring the alignment inside a tunnel. | CO1 | 6 |
| B) Write a note on Principle of EDM. | CO2 | 6 |
| C) The following perpendicular offsets were taken at 10m intervals from a survey line to an irregular boundary line 3.82, 4.37, 6.82, 5.26, 7.59, 8.90, 9.52, 8.42 & 6.43m. Calculate the area enclosed between survey line & boundary by-i) Simpson's Rule ii) Trapezoidal Rule | CO3 | 6 |

***** End *****

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2023

Course: B. Tech.

Branch: Civil Engineering

Semester: III

Subject Code & Name: BTCVC305 Surveying

Max Marks: 60

Date: 21/08/2023

Duration: 3 Hr.

Instructions to the Students:

- 1. All the questions are compulsory.*
- 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*
- 3. Use of non-programmable scientific calculators is allowed.*
- 4. Assume suitable data wherever necessary and mention it clearly.*

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Explain ranging and chaining in detail.	CO1	6
B) Illustrate methods of setting out perpendicular.	CO1	6
C) Discuss principles of surveying in detail.	CO1	6
Q.2 Solve Any Two of the following.		12
A) Explain various bearing systems.	CO1	6
B) Discuss surveyors compass with neat diagram.	CO1	6
C) Explain intersection method of Plane table surveying.	CO2	6
Q. 3 Solve Any Two of the following.		12
A) The following readings were taken with level 3.865, 3.345, 2.930, 1.950, 0.855, 3.795, 2.640, 1.540. The level was shifted after the 5 th reading. The first reading was taken on BM of 150.250. Calculate the reduced levels and apply usual checks.	CO3	6
B) Explain contouring in detail.	CO3	6
C) Illustrate different types of levelling.	CO3	6
Q.4 Solve Any Two of the following.		12
A) Explain theodolite with neat diagram.	CO3	6
B) Illustrate applications of theodolite in surveying.	CO3	6
C) Explain temporary and permanent adjustments of theodolite.	CO3	6
Q. 5 Solve Any Two of the following.		12
A) Explain various types of field surveys.	CO3	6
B) Illustrate curve ranging.	CO3	6
C) Discuss correlation of underground and surface surveys.	CO3	6

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular and Supplementary Winter Examination – 2023

Course: B. Tech. Branch: Civil & Allied Engineering Semester: III

Subject Code & Name: BTCIC305/BTCEC305/BTCVC305 Surveying

Max Marks: 60

Date: 11/01/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Define surveying. Write down applications of surveying. State Principle of Surveying.	CO1	6
B) Explain different methods of linear measurement.	CO1	6
C) Differentiate between plane surveying and geodetic surveying.	CO1	6
Q.2 Solve Any Two of the following.		12
A) Explain the various accessories used in plane table surveying with neat labelled diagram.	CO2	6
B) Define the following: (any three) a. Open traverse and closed traverse b. Fore bearing and Back bearing c. True meridian and Magnetic Meridian d. True bearing and Magnetic meridian	CO2	6
C) Convert the following WCB's to QB's a. WCB of AB = $45^{\circ}30'$ b. WCB of CD = $108^{\circ}45'$ c. WCB of GH = $75^{\circ}50'$ d. WCB of HK = $145^{\circ}20'$ e. WCB of ML = $340^{\circ}10'$ f. WCB of DE = $60^{\circ}10'$	CO2	6
Q. 3 Solve Any Two of the following.		12
A) Write down the characteristics of contour lines.	CO3	6
B) The following staff readings were taken successfully with level, the instrument having been move forward after the second, fourth and eighth reading:	CO3	6

0.0875, 1.235, 2.310, 1.385, 2.930, 3.125, 4.125, 0.120, 1.875, 2.030, 3.765

The first reading was taken on a benchmark of elevation 132.135 Enter the reading in level book form and reduced level. Calculate the reduced level by rise and fall method.

- | | | |
|---|------------|----------|
| C) Enlist the various types of leveling. Explain any one with neat labelled diagram. | CO3 | 6 |
|---|------------|----------|

Q.4 Solve Any Two of the following.	12
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|--|------------|----------|
| A) State the functions of following parts of Theodolite: 1. Eyepiece 2. Bubble tube 3. Lower Tangent screw 4. Index Frame 5. Telescope | CO4 | 6 |
| B) What are the different methods of traversing? Explain any one in detail. | CO4 | 6 |
| C) Define (any three):
1. Transiting of Theodolite 4. Telescope Normal
2. Swinging 5. Face left
3. Line of Collimation | CO4 | 6 |

Q. 5 Solve Any Two of the following.	12
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|--|------------|----------|
| A) Explain project survey on Railway Engineering. | CO4 | 6 |
| B) Write a Short note on Mine Surveying. | CO4 | 6 |
| C) Explain the steps involved while carrying out Engineering Survey | CO1 | 6 |

***** End *****

Supplementary Summer Examination – 2024

Course: B. Tech. Branch: Civil / Civil & Infra. / Civil & Environmental Engineering

Subject Code & Name: BTCEC305 / BTCVC305 / BTCIC305 Surveying

Max Marks: 60

Date: 09/07/2024

Semester: III

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Define surveying. Write down applications of surveying. State the principles of Surveying.	CO1	6
B) What is meant by Ranging? Explain the methods of ranging in detail.	CO1	6
C) Explain in detail Errors in Chaining.	CO1	6
Q.2 Solve Any Two of the following.		12
A) Define the following terms: i) True Meridian ii) Magnetic Bearing iii) Magnetic Meridian iv) True Bearing v) Arbitrary Meridian vi) Local Attraction	CO2	6
B) Convert the following WCB's to QB's a. WCB of AB = $45^{\circ}30'$ b. WCB of CD = $108^{\circ}45'$ c. WCB of GH = $75^{\circ}50'$ d. WCB of HK = $145^{\circ}20'$ e. WCB of ML = $340^{\circ}10'$ f. WCB of DE = $60^{\circ}10'$	CO2	6
C) Enlist the different methods of Plane Table and Explain any two in detail.	CO2	6
Q. 3 Solve Any Two of the following.		12
A) Write down the characteristics of contour lines.	CO3	6
B) During a levelling work started from a BM of known R.L.=100m. The following staff readings were obtained: 0.850, 1.555, 1.725, 0.455, 1.800, 1.750, 0.950, 1.555 The instrument was shifted after 3rd and 6th readings. Enter correctly all the readings in a field book page and determine the reduced levels of all stations showing calculations by Rise and Fall method. Apply usual arithmetic check.	CO3	6

C) What are the types of levelling operations explain with neat sketch. CO3 6

Q.4 Solve Any Two of the following. 12

A) What are the different methods of traversing Explain any one in detail. CO4 6

B) Explain Bowditch Rule and Transit Rule. CO4 6

C) Define (any three): CO4 6

1. Transiting of Theodolite
2. Swinging
3. Line of Collimation
4. Telescope Normal
5. Face left
6. Face right

Q. 5 Solve Any Two of the following. 12

A) Explain the Steps involved in survey of Waterways. CO4 6

B) Explain the steps involved while carrying out Engineering Survey. CO4 6

C) Write a short note on Reconnaissance. CO1 6

*** End ***